

ArduPilot Implementation of Optimal Control for Tilt Rotor UAV Landing

A thesis submitted in part fulfilment of the degree of
Bachelor of Engineering (Honours)

by
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This thesis contains no material which has been accepted for the award of any other degree or diploma in any university. To the best of the author's knowledge, it contains no material previously published or written by another person, except where due reference is made in the text.

Lincoln Karavas
3 October 2026

Acknowledgements

Thanks to my cat, Fluffy and my budgie, Rex.

Abstract

The abstract should be brief (100–150 words) and directed to readers familiar with the broad area of the thesis subject. “The abstract is of utmost importance for it is read by 10 to 500 times more people than hear or read the entire article. It should not be a mere recital of the subjects covered, replete with such expressions as “is discussed” and “is described.” It should be a condensation and concentration of the essential qualities of the paper.”

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List of Figures

1.1 Example figure. 1

List of Tables

4.1 Table captions are above tables. Tables are to be centered on the page. 5

Nomenclature

SOS

Save Our Souls

Introduction

The introduction orients the reader to your thesis. It should present the nature and scope of the problem in terms of applications to ‘real world’ problems (what motivates the study); refer briefly to associated and relevant literature already published; state the method of investigation (the main technical issues, including important limitations in scope); and describe the report that follows (a sort of road-map, so that the reader knows what’s to come). This is well shown in [1].



Figure 1.1: Example figure.

Literature Review

Methodology

Note that methodology varies according to project type. Consider the engineering tools you've learned throughout your degree in ANU to generate your results — this may also include the systems engineering approach.

3.1 Modelling

3.1.1 Aerodynamic Surfaces

3.1.2 Rotor Forces

3.1.3 Dynamics

3.2 Trajectory Optimisation

The matlab library OptimTrajectory `**intext**` was used to solve the optimisation problem for the transition trajectory. OptimTrajectory utilises matlab's fmincon solver.

3.2.1 Model Setup

The aircraft model discussed above was arranged into a statespace representation.

$$Y = \begin{bmatrix} x & z & \theta & \dot{x} & \dot{z} & \dot{\theta} \end{bmatrix}^T, U = \begin{bmatrix} \omega_f & \omega_r & \epsilon & \delta_e \end{bmatrix}^T$$

3.2.2 Constraints

The constraints for the trajectory optimisation were as follows.

$$Y_{initial} = \begin{bmatrix} -400 & -50 & 0 & * & * & 0 & 0 & 0 \end{bmatrix}^T$$

$$Y_{final} = \begin{bmatrix} 0 & -10 & 0 & 0 & 0 & 0 \end{bmatrix}^T$$

$$Y_{upper} = \begin{bmatrix} 0 & -10 & \pi/2 & 30 & \infty & \infty \end{bmatrix}^T$$

3.2.3 Initialisation**3.2.4 Mesh Refinement****3.2.5 Solution****3.3 Control****3.3.1 LQR****3.3.2 Lateral Stabilistaion****3.3.3 Hover****3.3.4 Level Flight****3.3.5 Landing Approach****3.4 Simulation Enviroment****3.4.1 Rotor Model****3.4.2 Safety Features****3.4.3 Altitude Floor****3.4.4 State Error Disconnect**

state error explanation goes here

3.5

Results and Analysis

When presenting your results, it is important that you organise them in an orderly manner to avoid clusters of information. Be critical and properly analyse what the results mean. Provide insightful discussion and explanation. Consider using other published articles to support or argue your results. Make sure it clear which work are yours and which are background materials. Also, include the implication of your results.

4.1 Experimental Results

4.2 Subheading 1

How to reference Figure 4.1.

Table 4.1: Table captions are above tables. Tables are to be centered on the page.

No	Student	Task
1	A	X
2	B	Y
3	C	Z

4.3 Analysis and Discussion

4.4 Subheading 2

4.4.1 How to do math equations

The well known Pythagorean theorem $x^2 + y^2 = z^2$ was proved to be invalid for other exponents. Meaning equation 4.1 has no integer solutions:

$$x^n + y^n = z^n$$

$$x^n + y^n = z^n \tag{4.1}$$

Conclusions and Future Development

Provide an answer to your research question or evaluation of overall results given. Discuss what worked well and did not work well in your research. Outline the limitations of approach, methodology, data or research. Give suggestions/recommendations for future work.

Appendix A: The First

Manage your references and present them here in order.

Appendix B: The Second

Manage your references and present them here in order.

Bibliography

- [1] S. Xu, K. Blackmore, and H. Jones. An analysis framework for mobility metrics in mobile ad hoc networks. *EURASIP Journal on Wireless Communications and Networking*, 2007:16, 2007.